ISHAN BHARDWAJ

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Quantitative finance graduate student with experience building end-to-end trading systems, predictive ML models, and automated analytical pipelines on large-scale financial datasets. Skilled in systematic strategy development, backtesting engine design, walk-forward validation, live broker integration (IBKR API), statistical modeling, derivatives pricing, and AI model evaluation. Targeting quant research, systematic trading, and data science in finance roles.

EDUCATION

University of Minnesota — Twin Cities M.S. Financial Mathematics — Expected Dec 2026

KIIT University B.Tech, Computer Science — Jul 2020 – May 2024

SKILLS

Trading & Quant Systematic strategy development, backtesting engine design, walk-forward IS/OOS validation, IBKR API (ib_insync), ES/NQ/CL futures, options pricing (Black-Scholes, binomial, Monte Carlo), Greeks, implied volatility

ML & Data Python (pandas, numpy, scikit-learn, XGBoost, TensorFlow/Keras), SHAP, pyarrow, asyncio, time-series forecasting, feature engineering, SQL-style data processing

Finance DCF valuation, SOTP, scenario analysis, equity research, credit risk modeling (Merton structural, Altman Z-Score), cointegration analysis, VaR/CVaR, KPI dashboards, Excel (advanced)

WORK EXPERIENCE

Quantitative Finance Expert (Contractor) — Jul 2026 – Present

Mercor

- Engaged through Mercor's selective expert vetting process to evaluate AI model outputs on graduate-level quantitative finance tasks — derivatives pricing (Black-Scholes, binomial trees, Monte Carlo), options Greeks, structural credit models (Merton), and systematic trading logic — providing authoritative feedback on mathematical accuracy and reasoning quality.
- Design ground-truth benchmark solutions for complex financial mathematics problems including stochastic processes, risk model construction (VaR/CVaR/Sharpe), and portfolio optimization; outputs used directly in training and evaluation of frontier AI systems.
- Identify model failure modes in high-stakes financial domains — flawed assumptions, computational errors, incorrect financial intuitions — improving AI reliability on quantitative reasoning tasks requiring M.S.-level finance expertise.

Equity Research Analyst Intern — Jun 2025 – Aug 2025

Atlasio Capital Partners

- Built predictive financial models across 40+ companies, analyzing revenue, cost structures, and leverage ratios to identify margin inefficiencies of 300–500 bps across target sectors.
- Designed automated data workflows integrating financial datasets with validation pipelines, reducing manual data inconsistencies by 25% and improving downstream model reliability.
- Developed scenario-based pricing and cost sensitivity models across 10+ assumptions, quantifying tradeoffs between growth, margins, and credit risk; Python pipeline automation improved research turnaround by 20%.

Quantitative Analyst Intern — Jan 2024 – Mar 2024

Mudraksh & McShaw

- Analyzed datasets exceeding 500,000 records to identify statistical trends, cost drivers, and performance anomalies impacting business outcomes across multiple product lines.
- Built automated data pipelines and workflow triggers reducing manual processing time by 20–25%; developed KPI dashboards tracking 15+ operational and financial metrics for management reporting.

Trade Support Analyst Intern — Nov 2023 – Dec 2023

AlgoBulls

- Processed and validated 1,000+ daily transactions across financial systems, ensuring near 100% accuracy and identifying discrepancies in transaction and pricing data to improve reporting reliability.
- Supported structured data workflows across 10,000+ transactions enabling audit readiness and regulatory compliance; contributed to automation initiatives reducing manual handling by 10–15%.

PROJECTS

CME Systematic Futures Trading System — CME Institute Competition, June 2026 — github.com/ISHAN217/CME-Futures-Trading-System

Python 3.12 · pandas · numpy · pyarrow · asyncio · IBKR API (ib_insync) · Custom bar-level backtesting engine

- Built an end-to-end automated futures trading system covering IBKR live data acquisition, bar-level backtesting with fill verification, and live signal generation across 5 independent strategies; achieved OOS Sharpe of 4.85, IS/OOS ratio of 1.34x, and profitability in all 5 held-out years (2022–2026).

- Conducted fill validity audit revealing 70–80% of backtested entries were unfillable in practice and corrected win rate from 79.9% to an honest 53–60%; built 38-check test suite verifying P&L to the cent, IS/OOS ratio, and year-by-year profitability.
- Explored 15 signal types across ES, NQ, and CL futures; rejected 11 after OOS failure with documented reasons, retaining only signals with positive Sharpe across all held-out years.

Options Pricing Engine — From Scratch, No QuantLib github.com/ISHAN217/options-pricing-engine

Python · Black-Scholes · CRR Binomial · Monte Carlo · Brent/Newton IV Solvers · Delta-Hedging Simulation

- Built three independent pricers (Black-Scholes analytical, CRR binomial tree with American early exercise, Monte Carlo with antithetic variates for European/Asian/barrier options) with all 5 Greeks verified against finite-difference to <1e-4; 19/19 test suite passing.
- Implied volatility solvers (Brent + Newton-Raphson) achieving round-trip accuracy <1e-6; live IV surface fitted from real SPY/AAPL options chain with OTM-only liquidity filtering. Delta-hedging P&L simulation validates vol carry formula across 1,000 paths.

Credit Default Model — Merton vs Altman vs XGBoost github.com/ISHAN217/credit-default-model

Python · Merton (1974) Structural Model · Altman Z-Score · XGBoost · SHAP TreeExplainer · 28 Financial Ratio Features

- Merton structural model (equity = call on firm assets) solved via Newton-Raphson; AUC 0.956 / KS 0.917 on 46-company universe — outperforming Altman (0.892) and XGBoost (0.778). SHAP confirmed log(market cap) and equity vol as top predictors, consistent with structural model theory.

Cointegration & Kalman Filter Pairs Trading github.com/ISHAN217/stat-arb-pairs-trading

Python · Engle-Granger · Johansen Cointegration · Kalman Filter · Walk-Forward IS/OOS Backtest · Hurst Exponent

- End-to-end pairs trading framework screening 190 pairs across 5 sectors; Kalman filter for dynamic hedge ratio vs rolling OLS. All 8 IS-cointegrated pairs had negative OOS Sharpe — documenting cointegration breakdown across 2021–2026 regime changes as the central finding.

MISO Day-Ahead LMP Forecasting — Directed Research, University of Minnesota github.com/ISHAN217/MISO-LMP-Forecasting

Python · scikit-learn · XGBoost · SHAP · pandas · matplotlib · seaborn

- ML forecasting pipeline for day-ahead electricity price prediction in MISO LRZ4; XGBoost/Ridge achieved RMSE $11.56/MWh and 75–76% directional accuracy vs 36% naive baseline — a 2x improvement. SHAP confirmed net load as strongest predictor.

MP Materials (NYSE: MP) — Strategic Equity Analysis | Airline Health Score (AHS+)

DCF · Scenario Analysis · Peer Comps · Rare Earth Markets | Python · DOT Data (DB1B, T-100, Form 41) · Statistical Modeling

- MP Materials: 12-month price target of $42.50 from $22.01 (+93%) achieved ahead of schedule via DoD procurement milestone; scenario-based DCF covering NdPr supply deficit, vertical integration, and EV/wind/defense demand.
- AHS+: fuel-neutral composite scoring system for 260+ U.S. carriers across 100 months of DOT data (2017–2025); ranking consistency rho = 0.9997, applied to equity research screening and operational benchmarking.